

# **Overview of Cables for Networking, Communication, and Data Systems**

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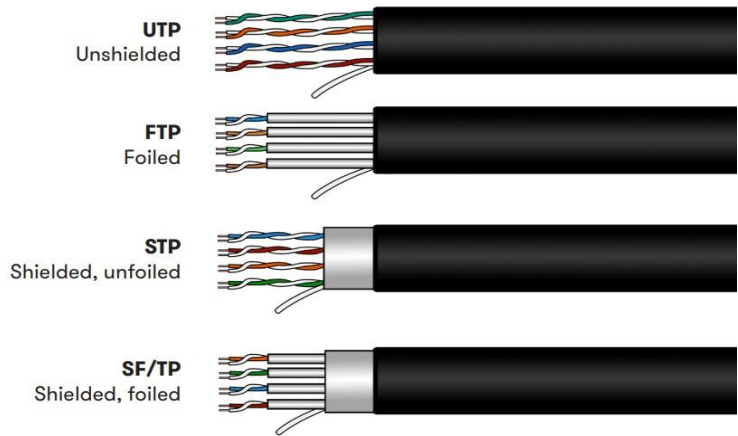
### **Conclusion**

❑ 1. Twisted Pair Cables

➤ Types & Categories

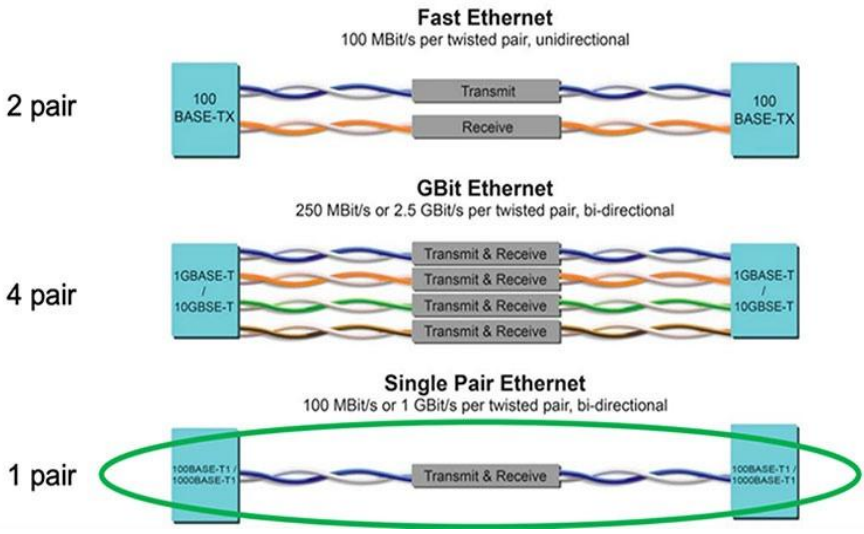
- ✓ UTP (Unshielded Twisted Pair): Most common, cost-effective, used in LAN networks.
- ✓ STP (Shielded Twisted Pair): Has shielding to reduce EMI, suitable for noisy environments.
- ✓ FTP (Foiled Twisted Pair): Each pair wrapped in foil for better interference protection.
- ✓ SF/TP (Shielded Foiled Twisted Pair): Combines overall shielding + individual pair shielding for high-performance networks.

Twisted pair (TP) screening



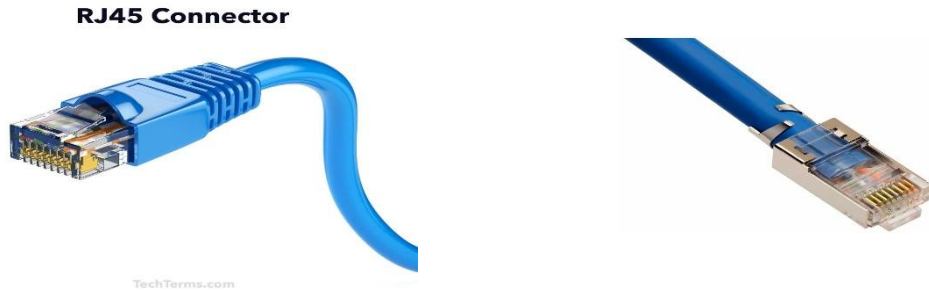
❑ 1.2 Speed & Bandwidth

| Cable Type | Speed     | Bandwidth | Distance                                                |
|------------|-----------|-----------|---------------------------------------------------------|
| Cat6       | 1–10 Gbps | 250 MHz   | Suitable for modern office LANs, up to 55m for 10 Gbps  |
| Cat6a      | 10 Gbps   | 500 MHz   | Supports full 10G over 100m, shielded against EMI       |
| Cat7       | 10 Gbps   | 600 MHz   | Fully shielded, ideal for industrial and high EMI areas |



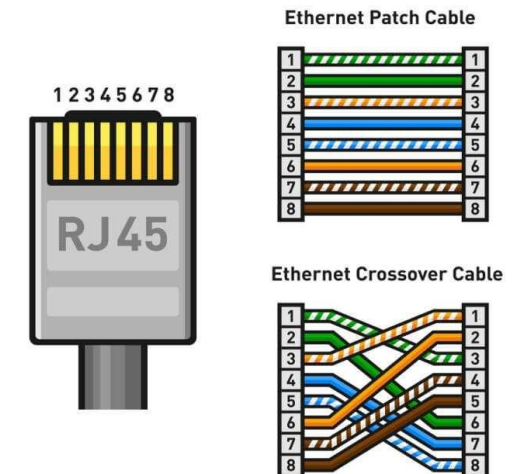
## ❑ 1.4 Connectors

- ✓ **RJ45 (8P8C):** Standard connector for UTP, Cat6, Cat6a, and Cat7 cables; easy to install.
- ✓ **Shielded RJ45:** Used with STP, FTP, and S/FTP cables; provides EMI protection for industrial or high-noise environments.
- ✓ **GG45 (GigaGate 45):** High-performance connector for Cat7/Cat7a; supports 10 Gbps and above; backward compatible with RJ45; better crosstalk and EMI protection.
- ✓ **Tera Connector:** Designed for Cat7/Cat7a; supports up to 40 Gbps; fully shielded; ideal for data centers; non-RJ45 form factor.



## ❑ Twisted Pair Cable Types & Color Code

| Features        | Straight-Through Cable                          | Crossover Cable                                     |
|-----------------|-------------------------------------------------|-----------------------------------------------------|
| Purpose         | Connects different devices (PC → Switch/Router) | Connects similar devices (PC → PC, Switch → Switch) |
| Wiring Standard | Same on both ends (568A–568A or 568B–568B)      | Different on each end (568A–568B)                   |
| Color Code      | Follow 568A or 568B                             | Follow 568A on one end, 568B on the other           |



## ❑ 1.5 Advantages & Disadvantages

### ➤ **Advantages:**

- ✓ Easy to install, widely available, and cost-effective.
- ✓ Supports Power over Ethernet (PoE) and is flexible for office and industrial LANs.

### ➤ **Disadvantages:**

- ✓ Limited distance for high-speed transmission (100 m max for standard Ethernet).
- ✓ Susceptible to EMI and crosstalk, especially unshielded cables.

## ❑ 2. Fiber Optic Cables

### ➤ **2.1 Types & Connectors**

#### ✓ **Single Mode Fiber (SMF):**

- ❖ Used for long-distance communication.
- ❖ Common connectors: LC, SC, ST, FC.

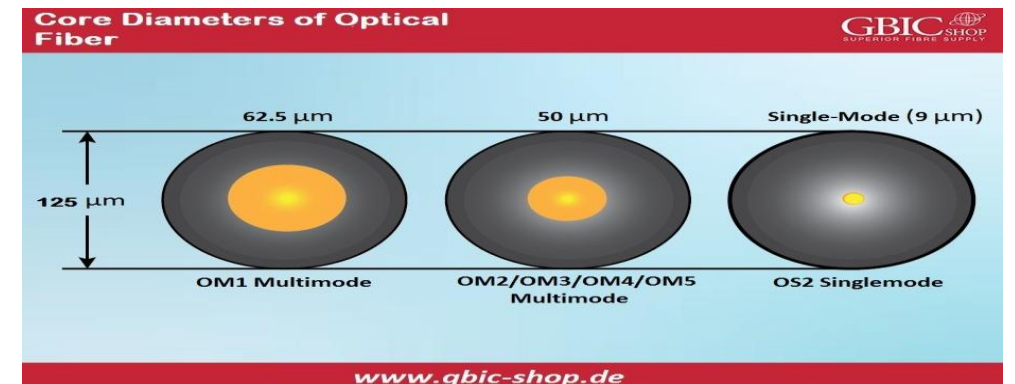
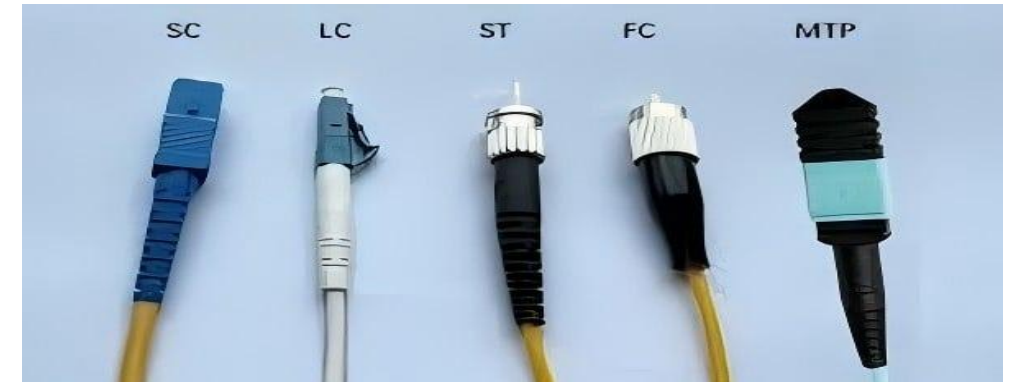
#### ✓ **Multimode Fiber (MMF):**

- ❖ Used for short-distance communication (LAN, industrial networks).
- ❖ Common connectors: LC, SC, ST, MPO/MTP.

## ❑ 2.2 Core Size & Connectors

- ✓ **SMF Core Size:** 9  $\mu\text{m}$ .
- ✓ **MMF Core Size:** 50  $\mu\text{m}$  or 62.5  $\mu\text{m}$ .

Connectors: LC, SC, ST, FC, MPO/MTP



❑ 2.3 Speed & Bandwidth

- ✓ **SMF:** Supports 10 Gbps – 400 Gbps, very high bandwidth.
- ✓ **MMF:** Supports 1 Gbps – 100 Gbps, suitable for LAN/backbone.



❑ 2.4 Distance Range

- ✓ **SMF:** Kilometers range, ideal for long-distance telecom and industrial links.
- ✓ **MMF:** Up to 550 m for 10 Gbps; shorter distances for higher speeds.



❑ 2.5 TYPES OF SFP MODULES

**SFP (Small Form-Factor Pluggable)** is a compact, hot-swappable module used in switches, routers, OLT/ONT, relays, and media converters to connect fiber optic or copper cables.

Allows easy change of port type (SM/MM/Copper) without replacing the device.

| SFP Type      | Speed        | Cable Type         | Connector | Usage                                  |
|---------------|--------------|--------------------|-----------|----------------------------------------|
| SFP (1G)      | Up to 1 Gbps | SMF / MMF / Copper | LC / RJ45 | Basic network links, switches, routers |
| SFP+ (10G)    | 10 Gbps      | SMF / MMF          | LC        | High-speed backbone, substation LAN    |
| SFP28 (25G)   | 25 Gbps      | SMF / MMF          | LC        | Modern switches, medium backbone       |
| QSFP (40G)    | 40 Gbps      | SMF / MMF          | MPO/MTP   | Data centers, high bandwidth links     |
| QSFP+ (40G )  | 40 Gbps      | SMF / MMF          | MPO/MTP   | High-performance networks              |
| QSFP28 (100G) | 100 Gbps     | SMF / MMF          | MPO/MTP   | Core networks, large substations       |

## ❑ 2.5 Advantages & Disadvantages

### ➤ **Advantages:**

- ✓ Immune to EMI and crosstalk.
- ✓ Supports very high bandwidth and long distances.

### ➤ **Disadvantages:**

- ✓ More expensive than twisted pair cables.
- ✓ Fragile, requires careful installation and termination.

## ❑ 3. Coaxial Cables (GPS in Substation)

### ➤ **3.1 Types**

- ✓ **RG-58:** Used for short-distance GPS and communication signals.
- ✓ **RG-6:** Low-loss cable used for longer GPS antenna runs.
- ✓ **LMR-400:** Very low-loss coaxial cable used for long GPS cable lengths in substations.

## ❑ 3.2 Connectors

- ✓ **BNC Connector:** Most common for GPS receivers and clocks.
- ✓ **N-Type Connector:** Used for outdoor GPS antennas; weatherproof and low-loss.
- ✓ **TNC Connector:** Threaded version of BNC, good for vibration environments.

| COAXIAL CONNECTOR                                                                                                                                                                                                                    | IMAGE                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>RP-TNC</b><br>A derivative of a TNC connector, the RP-TNC connector is one of the most frequently used cable connectors for a UHF RFID system.                                                                                    | <br>RP-TNC MALE   |
| <b>SMA</b><br>SMA connectors are known for their small size in comparison to other typical UHF connectors, and are about the size of a pencil eraser.                                                                                | <br>SMA MALE      |
| <b>N-TYPE</b><br>N-TYPE connectors are almost twice as big as an RP-TNC connector, so they are the largest connectors commonly used in UHF RFID systems.                                                                             | <br>N-TYPE MALE   |
| <b>TNC</b><br>A relative of the RP-TNC, the TNC connector is the normal polarity version.                                                                                                                                            | <br>TNC MALE      |
| <b>RP-SMA</b><br>A derivative of an SMA connector, an RP-SMA is simply an SMA connector with the polarity reversed.                                                                                                                  | <br>RP-SMA MALE |
| <b>BNC</b><br>BNC is a less commonly used connector in UHF RFID systems, but is similar in size to a TNC connector. BNC connectors allow for quicker attachment than most others, but they also are more likely to loosen over time. | <br>BNC MALE    |



### ❑ 3.3 Performance (Speed/Bandwidth)

- ✓ Coaxial cables **do not carry data speed like Ethernet**; they carry **GPS RF signals** (1.57542 GHz – L1 band).
- ✓ Designed for **low-loss and stable frequency transmission** rather than Mbps/Gbps speeds.

### ❑ 3.4 Distance

- ✓ **RG-58**: Up to ~20–25 m (beyond this, high loss).
- ✓ **RG-6**: 30–40 m.
- ✓ **LMR-400**: Up to 70–80 m with very low signal loss.

### ❑ 3.5 Advantages & Disadvantages

#### ➤ **Advantages:**

- ✓ Very low signal loss for GPS frequency.
- ✓ Strong shielding protects against EMI in substations.

#### ➤ **Disadvantages:**

- ✓ Limited bending flexibility for thicker cables (LMR-400).
- ✓ Longer runs require thicker and more expensive coax.



## ❑ 4. Serial Communication Cables

### ➤ 4.1 Types

- ✓ RS-232 Cable: Used for point-to-point device communication (PC → Relay/BCU).
- ✓ RS-485 Cable: Used for multi-drop networks (SCADA, Modbus, RTU).

### ❑ 4.2 Connectors

- ✓ DB9 (DE-9): Most common serial connector for relays, IEDs, and PCs.
- ✓ DB25: Older connector used in legacy equipment.

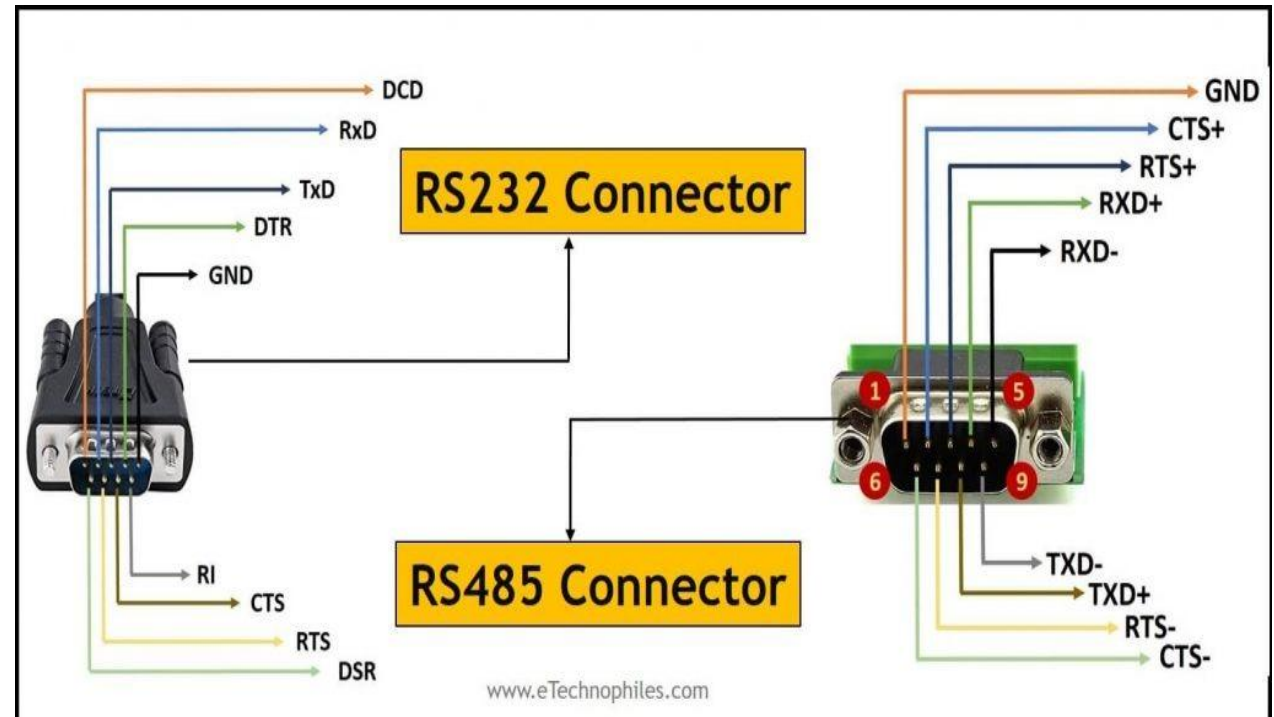
## ❑ 4.3 Speed & Distance

### ➤ RS-232

- ✓ Speed: Up to 115.2 kbps
- ✓ Distance: Max 15 m

### ➤ RS-485

- ✓ Speed: Up to 10 Mbps
- ✓ Distance: Up to 1200 m (depends on baud rate)



## ❑ 4.4 Advantages & Disadvantages

### ➤ **Advantages:**

- ✓ Simple, reliable communication for field devices.
- ✓ RS-485 supports long distance and multi-node networks.

### ➤ **Disadvantages:**

- ✓ Low data speed compared to Ethernet.
- ✓ Limited device compatibility and requires proper termination.

## ❑ 5. Data & Peripheral Cables

### ➤ **5.1 Types**

- ✓ USB Cables (USB 2.0 / 3.0 / 3.1 / Type-C) – Used for data transfer and peripherals.
- ✓ HDMI Cable – Used for display and audio.
- ✓ DisplayPort Cable – High-resolution display interface.
- ✓ VGA Cable – Legacy analog display cable.
- ✓ PS/2 Cable – Used for old keyboard/mouse connections.

| Quick Facts                  |                                                                                      |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| DisplayPort/HDMI comparison. |                                                                                      |                                                                                      |
|                              | DisplayPort                                                                          | HDMI                                                                                 |
|                              |  |  |
| Cable length                 | 2 m; Reduced bandwidth: 15 m                                                         | 5m                                                                                   |
| Top bandwidth                | 21.6 Gbps                                                                            | 10.2 Gbps                                                                            |
| Daisychain monitors          | 4                                                                                    | None                                                                                 |
| Locking connector            | Yes                                                                                  | No                                                                                   |
| Backward compatibility       | HDMI, DVI, VGA with adapter                                                          | DVI                                                                                  |
| Standard                     | Open                                                                                 | Licensed                                                                             |
| Market                       | PC world                                                                             | Consumer electronics                                                                 |

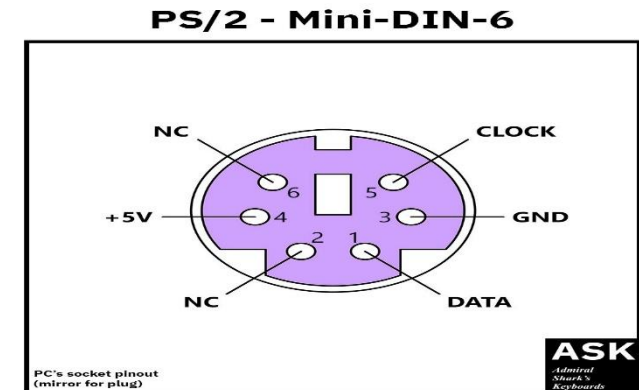
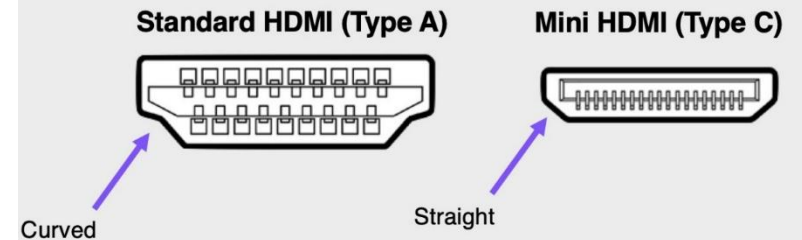
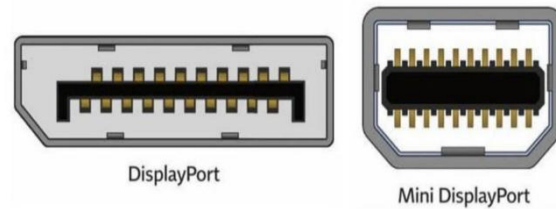
## ❑ 5.2 Connectors

- ❖ USB: Type-A, Type-B, Mini USB, Type-C.
- ❖ HDMI: Standard HDMI, Mini HDMI.
- ❖ DisplayPort: DP standard, Mini DP.
- ❖ VGA: 15-pin D-Sub connector.
- ❖ PS/2: 6-pin mini-DIN connector.



## ❑ 5.3 Speed & Bandwidth

- ❖ **USB 2.0:** 480 Mbps
- ❖ **USB 3.0:** 5 Gbps
- ❖ **USB 3.1/3.2:** 10–20 Gbps
- ❖ **HDMI:** Up to 48 Gbps (HDMI 2.1)
- ❖ **DisplayPort:** Up to 80 Gbps (DP 2.0)



## ❑ 5.4 Distance

- ❖ **USB:** ~5 m (can extend with active cable).
- ❖ **HDMI:** 10–15 m (beyond 10 m needs booster).
- ❖ **DisplayPort:** 2–3 m for high resolutions; up to 15 m for lower resolutions.
- ❖ **VGA:** Up to 30 m (quality drops with distance).

## ❑ 5.5 Advantages & Disadvantages

### ➤ **Advantages:**

- ✓ Supports wide range of devices (keyboard, mouse, displays, storage).
- ✓ High-speed data and display transmission options available.

### ➤ **Disadvantages:**

- ✓ Some cables (HDMI/DP) are distance-limited.
- ✓ Legacy cables (VGA, PS/2) offer low performance and compatibility issues.

## ❑ Conclusion

This documentation gives a clear overview of the main cables used in network, communication, and data systems. Knowing their types, connectors, speed, distance limits, and benefits helps in choosing the right cable for any setup. With the correct cable selection and installation, overall system performance, reliability, and communication quality improve.